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Assignment 1

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Question 1

Part (a)

$$\mathbf{Y} = \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix}.$$

$$\mathbf{Y}^2 = \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix} \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix} = \begin{bmatrix} \vec{y}(-\vec{y}^\dagger) & 0 \\ 0 & -\vec{y}^\dagger(\vec{y}) \end{bmatrix} = \begin{bmatrix} -|\vec{y}|^2 \mathbf{P} & 0 \\ 0 & -|\vec{y}|^2 \end{bmatrix} = -|\vec{y}|^2 \begin{bmatrix} \mathbf{P} & 0 \\ 0 & 1 \end{bmatrix}.$$

Even powers: $\mathbf{Y}^{2k} = \mathbf{Y}^{2k} = \begin{bmatrix} -|\vec{y}|^2 \mathbf{P} & 0 \\ 0 & -|\vec{y}|^2 \end{bmatrix}^k$

$$\begin{bmatrix} -|\vec{y}|^2 \mathbf{P} & 0 \\ 0 & -|\vec{y}|^2 \end{bmatrix}^k = \begin{bmatrix} (-|\vec{y}|^2 \mathbf{P})^k & 0 \\ 0 & (-|\vec{y}|^2)^k \end{bmatrix} = \begin{bmatrix} (-|\vec{y}|^2)^k \mathbf{P} & 0 \\ 0 & (-)^k |\vec{y}|^{2k} \end{bmatrix} = (-)^k |\vec{y}|^{2k} \begin{bmatrix} \mathbf{P} & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\mathbf{Y}^3 = \mathbf{Y}^2 \mathbf{Y} = \begin{bmatrix} -|\vec{y}|^2 \mathbf{P} & 0 \\ 0 & -|\vec{y}|^2 \end{bmatrix} \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{0}_n & -|\vec{y}|^2 \mathbf{P} \vec{y} \\ |\vec{y}|^2 \vec{y}^\dagger & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{0}_n & -|\vec{y}|^2 \vec{y} \\ |\vec{y}|^2 \vec{y}^\dagger & 0 \end{bmatrix} = -|\vec{y}|^2 \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix}.$$

$$\implies \mathbf{Y}^3 = -|\vec{y}|^2 \mathbf{Y}.$$

Odd powers: $\mathbf{Y}^{2k+1} = \mathbf{Y}^{2k} \mathbf{Y}$

$$\rightarrow (-)^k |\vec{y}|^{2k} \begin{bmatrix} \mathbf{P} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix} = (-)^k |\vec{y}|^{2k} \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix} = (-)^k |\vec{y}|^{2k} \mathbf{Y}.$$

$$e^{\mathbf{Y}} = \sum_{n=0}^{\infty} \frac{\mathbf{Y}^n}{n!}, \quad \mathbf{Y}^0 = I, \quad U \mathbf{Y} U^\dagger = 1, \quad \det \mathbf{Y} = 1.$$

$$e^{\mathbf{Y}} = \mathbf{1} + \frac{1}{-|\vec{y}|^2} \sum_{k=1}^{\infty} \frac{(-1)^k (|\vec{y}|)^{2k}}{(2k)!} \mathbf{Y}^2 + \sum_{k=0}^{\infty} \frac{(-1)^k (|\vec{y}|)^{2k}}{(2k+1)!} \mathbf{Y}.$$

$$\frac{1}{-|\vec{y}|^2} \sum_{k=1}^{\infty} \frac{(-1)^k (|\vec{y}|)^{2k}}{(2k)!} \mathbf{Y}^2 = \frac{1}{-|\vec{y}|^2} (\cos |\vec{y}| - 1) \mathbf{Y}^2, \quad \sum_{k=0}^{\infty} \frac{(-1)^k (|\vec{y}|)^{2k}}{(2k+1)!} \mathbf{Y} = \frac{\sin |\vec{y}|}{|\vec{y}|} \mathbf{Y}.$$

$$\implies e^{\mathbf{Y}} = \mathbf{1} + \frac{1}{-|\vec{y}|^2} (\cos |\vec{y}| - 1) \mathbf{Y}^2 + \frac{\sin |\vec{y}|}{|\vec{y}|} \mathbf{Y} = \mathbf{1} + (\cos |\vec{y}| - 1) \begin{bmatrix} \mathbf{P} & 0 \\ 0 & 1 \end{bmatrix} + \frac{\sin |\vec{y}|}{|\vec{y}|} \begin{bmatrix} \mathbf{0}_n & \vec{y} \\ -\vec{y}^\dagger & 0 \end{bmatrix}.$$

$$\implies e^{\mathbf{Y}} = \begin{bmatrix} \mathbf{1}_n + (\cos |\vec{y}| - 1) \mathbf{P} & \sin |\vec{y}| \hat{y} \\ -\sin |\vec{y}| \hat{y}^\dagger & \cos |\vec{y}| \end{bmatrix}.$$

CHECK : $e^{\mathbf{Y}} e^{\mathbf{Y}^\dagger} = \mathbf{1}$ and $\det e^{\mathbf{Y}} = 1$

Part (b)

$$\mathbf{x} = \begin{bmatrix} \mathbf{1}_n + (\cos |\vec{y}| - 1) \mathbf{P} & \sin |\vec{y}| \hat{y} \\ -\sin |\vec{y}| \hat{y}^\dagger & \cos |\vec{y}| \end{bmatrix} \begin{bmatrix} \vec{0} \\ 1 \end{bmatrix} = [\sin |\vec{y}| \hat{y} \quad \cos |\vec{y}|]^T.$$

To show it lies on the hypersphere: we require

$$\sum_{i=1}^{2n+1} \|x_i\|^2 = 1, \quad \|\sin |\vec{y}| \hat{y}\|^2 + \|\cos |\vec{y}|\|^2 = 1, \quad \sin^2 |\vec{y}| + \cos^2 |\vec{y}| = 1.$$

Thus it lies on the hypersphere. Namely

$$\mathbf{x} \in S^{2n+1} \subset \mathbb{C}^{n+1} \cong \mathbb{R}^{2(n+1)}.$$

We can look at the dimension of $\mathbf{x} = [\sin |\vec{y}| \hat{y} \quad \cos |\vec{y}|]^T$:

$$\underbrace{\sin |\vec{y}| \hat{y}}_{\dim=2n} \quad \underbrace{\cos |\vec{y}|}_{\dim=1}.$$

Generally $\mathbf{x} \in \mathbb{R}^{2n+2}$ since $\mathbf{x}$ has $n+1$ complex components giving $2n+2$ real DOF. When we restrict $\mathbf{x}$ to lie on the hypersphere ($|\mathbf{x}| = 1$) then we lose a DOF ($\mathbf{x} \in \mathbb{S}^{2n+1}$) and we also notice that the last component ($\cos |\vec{y}|$) is strictly real. Geometrically, we restrict the values from a circle in the complex plane associated with that component of $\mathbf{x}$ to only take the values where that circle intersects the real axis. In realising that the last component is strictly real we mod out $U(1)$ symmetry. So we have ($\mathbf{x} \in \mathcal{M} \subset \frac{\mathbb{S}^{2n+1}}{U(1)}$). Notice that is the definition of the complex projected space $\mathbb{CP}^n \equiv \frac{\mathbb{S}^{2n+1}}{U(1)}$

Note:-

We can set $|\vec{y}| \in [0, \pi]$ as we can only determine $\vec{y}$ modulo 2π . $\hat{y}$ ranges over the whole sphere so if we set $|\vec{y}| \rightarrow 2\pi - |\vec{y}|$ it flips the sign of the first component.

$$\sin(2\pi - |\vec{y}|) = -\sin |\vec{y}|, \quad \cos(2\pi - |\vec{y}|) = \cos |\vec{y}|.$$

which is fine because we can simply let $\hat{y} \rightarrow -\hat{y}$. Thus we can restrict $|\vec{y}| \in [0, \pi]$ as this keeps $\sin |\vec{y}| \geq 0$ which avoids double counting.

The extra dimension in $\mathbb{S}^{2n+1}$ and not in $\mathcal{M}$ is those points in $\mathbb{S}^{2n+1}$ that differ by a phase. That is $\mathbf{x}$ and $e^{i\theta}\mathbf{x}$ are identified in $\mathcal{M}$. This is because the points in $\mathbb{S}^{2n+1}$ that differ by a phase correspond to the same physical state in quantum mechanics. Thus we have shown that $\mathcal{M} \cong \mathbb{S}^{2n+1}/U(1) = \mathbb{CP}^n$ which is the Hopf fibration.

Question 2

Part (a)

$$so(3, 1) = \left\{ X = \begin{bmatrix} A & C \\ C^T & B \end{bmatrix} \middle| A \in M_3(\mathbb{R}), B \in M(\mathbb{R}), C \in M_3(\mathbb{R}) \right\}.$$

We can define in terms of boosts $\beta = (\beta_1, \beta_2, \beta_3)$ and rotations $\theta = (\theta_1, \theta_2, \theta_3)$ the elements of $so(3, 1)$ as

$$\mathbf{X}(\beta, \theta) = \begin{bmatrix} 0 & \beta_x & \beta_y & \beta_z \\ \beta_x & 0 & \theta_z & -\theta_y \\ \beta_y & -\theta_z & 0 & \theta_x \\ -\beta_z & \theta_y & -\theta_x & 0 \end{bmatrix}.$$

$$K_x = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad K_y = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad K_z = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix},$$

$$J_x = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}, \quad J_y = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad J_z = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

where K_i are the boost generators and J_i are the rotation generators.

For $sl(2, \mathbb{C})$

$$sl(2, \mathbb{C}) = \left\{ X = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix} \in M_2(\mathbb{C}) \mid \text{tr } X = 0 \right\}.$$

We can choose the basis of $sl(2, \mathbb{C})$ to be the Pauli matrices

$$\sigma^1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma^2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma^3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

where they satisfy $\text{tr}(\sigma_i \sigma_j) = 2\delta_{ij}$ and under the Hilbert-Schmidt inner product $\langle A, B \rangle = \frac{1}{2} \text{tr}(AB^\dagger)$, $x_i = \frac{1}{2} \text{tr}(X \sigma_i)$ they are orthogonal. Thus we can express X as

$$2X = \sum_{i=1}^3 x_i \sigma^i, \quad x_i \in \mathbb{C}.$$

$$\implies 2X = \begin{bmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{bmatrix}.$$

We can compare to the definition above

$$\alpha = \frac{x_3}{2}, \quad \beta = \frac{x_1 - ix_2}{2}, \quad \gamma = \frac{x_1 + ix_2}{2}.$$

$$\implies x_1 = \beta + \gamma, \quad x_2 = i(\beta - \gamma), \quad x_3 = 2\alpha.$$

Thus $\forall x_i$ we can decompose it into real components $x_i = y_i + iz_i$ where $y_i, z_i \in \mathbb{R}$. Thus we have 6 real parameters which is the same as $so(3, 1)$.

Explicitly:

$$X = \frac{1}{2} \sum_{i=1}^3 (y_i + iz_i) \sigma^i = \frac{1}{2} (\mathbf{y} \cdot \boldsymbol{\sigma}) + \frac{i}{2} (\mathbf{z} \cdot \boldsymbol{\sigma})$$

Explicitly the real parameters for $sl(2, \mathbb{C})$ are y_1, y_2, y_3 and z_1, z_2, z_3 . We can write in terms of the following hermitian and anti-hermitian matrices

$$T = \frac{1}{2} \sigma^i, \quad S = \frac{i}{2} \sigma^i.$$

Explicitly:

$$T_x = \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad S_x = \frac{i}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$T_y = \frac{1}{2} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad S_y = \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$T_z = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad S_z = \frac{i}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Thus $\forall X \in sl(2, \mathbb{C})$ we can write it as

$$X = \sum_{i=1}^3 (y_i T_i + z_i S_i), \quad y_i, z_i \in \mathbb{R}$$

Part (b)

$\mathfrak{sl}(2, \mathbb{C})$.

The standard commutation relations for pauli matrices are

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma^k.$$

Thus we can check the following

$$[T_i, T_j], \quad [T_i, S_j], \quad [S_i, S_j].$$

$$[T_i, T_j] = \frac{1}{4} [\sigma_i, \sigma_j] = \frac{i}{2} \epsilon_{ijk} \sigma^k = \epsilon_{ijk} S^k.$$

$$[T_i, S_j] = \frac{i}{4} [\sigma_i, \sigma_j] = -\frac{1}{2} \epsilon_{ijk} \sigma^k = -\epsilon_{ijk} T^k.$$

$$[S_i, S_j] = -\frac{1}{4} [\sigma_i, \sigma_j] = -\frac{i}{2} \epsilon_{ijk} \sigma^k = -\epsilon_{ijk} S^k.$$

Clearly if any combination has the same indices e.g $[T_i, S_i]$ they vanish by properties of the levita-chevita and we also have antisymmetry.

so(3, 1).

The non-zero commutation relations for $so(3, 1)$ are given by (*quoting ACM II, HW II*)

$$[L_i, L_j] = \epsilon_{ijk} L_k$$

$$[L_i, K_j] = \epsilon_{ijk} K_k$$

$$[K_i, K_j] = -\epsilon_{ijk} L_k.$$

Part (c)

Definition 0.0.1: Isomorphism :

Two lie groups are isomorphic if:

- Their underlying manifolds $\mathcal{M}$ are topologically equivalent: they can be smoothly deformed into each other. This requires their topological properties, such as dimension, connectedness, and compactness, etc to be the same.
- The functions defining the group composition laws ϕ are equivalent: there is a smooth change of variables that deforms one function into the other while preserving the group structure.

The algebras can be compared directly via their commutation relations.

$$[S_i, S_j] = -\epsilon_{ijk} S^k \quad [K_i, K_j] = -\epsilon_{ijk} L_k$$

$$[T_i, T_j] = \epsilon_{ijk} S^k \quad [L_i, L_j] = \epsilon_{ijk} L_k$$

$$[T_i, S_j] = -\epsilon_{ijk} T^k \quad [L_i, K_j] = \epsilon_{ijk} K_k$$

.

We have S_i, T_i for $sl(2, \mathbb{C})$ and K_i, L_i for $so(3, 1)$. Define $\chi : so(3, 1) \rightarrow sl(2, \mathbb{R})$

$$\text{Ansatz : } K_i \mapsto T_i \quad [T_i, T_j] = \epsilon_{ijk} S^k, \quad [K_i, K_j] = -\epsilon_{ijk} L_k \implies L_i \mapsto -S_i.$$

Verify the other commutation relations:

$$\begin{aligned}[L_i, L_j] &\mapsto [-S_i, -S_j] = -\epsilon_{ijk} S^k \\ [L_i, K_j] &\mapsto [-S_i, T_j] = -\epsilon_{ijk} T^k\end{aligned}$$

We have the commutation relations for both algebras match under the identification

$$\begin{aligned}L_i &\leftrightarrow -S_i = -\frac{i}{2}\sigma^i \\ K_i &\leftrightarrow T_i = \frac{1}{2}\sigma^i.\end{aligned}$$

Under this identification we can see that

$$so(3, 1) \cong sl(2, \mathbb{C})_{\mathbb{R}}.$$

Thus the linear map $\phi : so(3, 1) \rightarrow sl(2, \mathbb{C})$ defined by

$$\begin{aligned}\phi(K_i) &= T_i \\ \phi(L_i) &= S_i.\end{aligned}$$

is a Lie algebra isomorphism since it preserves the commutation relations.

Part (d)

Definition 0.0.2: Adjoint :

The adjoint representation of a Lie group G is a way to represent the elements of the group as linear transformations acting on its own Lie algebra .

The adjoint operation is defined as

$$\text{Ad}_g(X) = gXg^{-1}.$$

Here we can use

$$\text{ad}_X(Y) = [X, Y].$$

Using our commutation relations

$$\begin{aligned}[J_i, J_j] &= \epsilon_{ijk} J^k \\ [J_i, K_j] &= \epsilon_{ijk} K^k \\ [K_i, K_j] &= -\epsilon_{ijk} J^k\end{aligned}$$

Let $a_i, b_i \in \mathbb{R}$ then an arbitrary element of $so(3, 1)$ can be written as

$$X = a_i J_i + b_i K_i = \begin{bmatrix} 0 & b_1 & b_2 & b_3 \\ b_1 & 0 & a_2 & -a_1 \\ b_2 & -a_2 & 0 & a_1 \\ b_3 & a_1 & -a_1 & 0 \end{bmatrix}.$$

$$\begin{aligned}\text{ad}_X(J_j) &= [X, J_j] = a_i [J_i, J_j] + b_i [K_i, J_j] = a_i \epsilon_{ijk} J^k - b_i \epsilon_{ijk} K^k \\ \text{ad}_X(K_j) &= [X, K_j] = a_i [J_i, K_j] + b_i [K_i, K_j] = a_i \epsilon_{ijk} K^k - b_i \epsilon_{ijk} J^k\end{aligned}$$

For a_i and b_i we can define a matrix $\epsilon_{ijk}a_i = \xi(a)$ and $\epsilon_{ijk}b_i = \xi(b)$ where

$$\xi(v) = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}.$$

As we want the first three columns/rows to correspond to $\mathfrak{so}(\mathbf{3})$ we take the rotation part ($v_i = a_i$) to be the top left 3×3 block. The boosts corresponding to b_i encapsulate temporal aspects (time dialation) which contribute to the temporal part of $\mathfrak{so}(\mathbf{3}, \mathbf{1})$.

Thus we can write the adjoint representation as

$$\text{ad}_X = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ -\mathbf{B} & \mathbf{A} \end{bmatrix} = \begin{bmatrix} 0 & -a_3 & a_2 & 0 & -b_3 & b_2 \\ a_3 & 0 & -a_1 & b_3 & 0 & -b_1 \\ -a_2 & a_1 & 0 & -b_2 & b_1 & 0 \\ 0 & b_3 & -b_2 & 0 & -a_3 & a_2 \\ b_3 & 0 & b_1 & a_3 & 0 & -a_1 \\ b_2 & -b_1 & 0 & -a_2 & a_1 & 0 \end{bmatrix}.$$

Part (e)

$$\text{ad}_X^2 = \text{ad}_X \text{ad}_X = \begin{bmatrix} \mathbf{A}^2 - \mathbf{B}^2 & \mathbf{AB} + \mathbf{BA} \\ -\mathbf{AB} - \mathbf{BA} & \mathbf{A}^2 - \mathbf{B}^2 \end{bmatrix}.$$

$$\begin{aligned} \text{tr}(\text{adj}_X^2) &= 2 \text{tr}(\mathbf{A}^2 - \mathbf{B}^2) \\ &= 2(\text{tr}(\mathbf{A}^2) - \text{tr}(\mathbf{B}^2)) \end{aligned}$$

$$\mathbf{A}^2 = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} = \begin{bmatrix} -a_3^2 - a_2^2 & * & * \\ * & -a_3^2 - a_1^2 & * \\ * & * & -a_2^2 - a_1^2 \end{bmatrix}$$

$$\text{tr}(\mathbf{A}^2) = -2(a_1^2 + a_2^2 + a_3^2) = -2|\mathbf{a}|^2$$

$$\text{tr}(\mathbf{B}^2) = -2|\mathbf{b}|^2$$

$$\Rightarrow \text{tr}(\text{adj}_X^2) = -4(|\mathbf{a}|^2 - |\mathbf{b}|^2) = 4|\mathbf{b}|^2 - 4|\mathbf{a}|^2 = K(X, X).$$

CHECK VIA INDEX NOTATION

Question 3

Part (a)

Given

$$\text{tr}(\lambda_I \lambda_J) = 2\delta_{IJ} \tag{1}$$

$$[\lambda_I, \lambda_J] = 2if_{IJK}\lambda^K \tag{2}$$

$$\{\lambda_I, \lambda_J\} = \rho\delta_{IJ}\mathbf{1}_3 + 2d_{IJK}\lambda^K. \tag{3}$$

Take the trace of (3):

$$\text{tr}(\{\lambda_I, \lambda_J\}) = \text{tr}(\rho\delta_{IJ}\mathbf{1}_3 + 2d_{IJK}\lambda^K) = \text{tr}(\rho\delta_{IJ}\mathbf{1}_3) + 2d_{IJK}\text{tr}(\lambda^K).$$

By definition λ^K are traceless so the second term vanishes. Thus we have

$$\text{tr}(\{\lambda_I, \lambda_J\}) = 3\rho\delta_{IJ} \quad (4)$$

$$\text{tr}(\{\lambda_I, \lambda_J\}) = \text{tr}(\lambda_I\lambda_J + \lambda_J\lambda_I) = 2\text{tr}(\lambda_I\lambda_J) = 4\delta_{IJ} \quad (5)$$

Equating (5) and (6) gives

$$3\rho\delta_{IJ} = 4\delta_{IJ} \implies \rho = \frac{4}{3}.$$

Part (b)

$$\begin{aligned} ([\lambda_I, \lambda_J] \lambda_K) &= (2if_{IJL}\lambda^L\lambda_K) \\ \text{tr}([\lambda_I, \lambda_J] \lambda_K) &= \text{tr}(2if_{IJL}\lambda^L\lambda_K) = 2if_{IJK}2\delta_{LK} \\ f_{IJL} &= \frac{1}{2i}\frac{1}{2}\text{tr}([\lambda_I, \lambda_J] \lambda_L) = \frac{1}{4i}\text{tr}(\lambda_I\lambda_J\lambda_L - \lambda_J\lambda_I\lambda_L) \end{aligned}$$

To show antisymmetry $I \leftrightarrow J$:

$$f_{JIL} = \frac{1}{4i}\text{tr}([\lambda_J, \lambda_I] \lambda_L) = \frac{1}{4i}\text{tr}(\lambda_J\lambda_I\lambda_L - \lambda_I\lambda_J\lambda_L) = -f_{IJL}.$$

To find the non-zero components we can use index notation and check the upper triangular components only (equivalently we can only calculate the positive entries:

To find non-zero components we can look at the 2×2 subspaces $\mathcal{S}_{ij}$ that behave like $su(2)$ algebras comprising the 3×3 vector space the Gell-Mann matrices act on.

$$\begin{aligned} \mathcal{S}_{12} &= \text{span} \{ \lambda_1, \lambda_2, \lambda_3 \} \\ \mathcal{S}_{13} &= \text{span} \left\{ \lambda_4, \lambda_5, \frac{1}{2}(\lambda_3 + \sqrt{3}\lambda_8) \right\} \\ \mathcal{S}_{23} &= \text{span} \left\{ \lambda_6, \lambda_7, \frac{1}{2}(-\lambda_3 + \sqrt{3}\lambda_8) \right\} \end{aligned}$$

Within each subspace the Gell-Mann matrices satisfy the $su(2)$ commutation relations, i.e. they act like Pauli matrices.

The non-vanishing structure constants are from the commutator of the off diagonal elements producing the diagonal elements up to the corresponding structure constant or when the commutator of off diagonal elements from different $\mathcal{S}_{ij}$ subspaces produce off diagonal elements from the remaining $\mathcal{S}_{kl}$ subspace.

$$\begin{aligned} \lambda_1 &= E_{12} + E_{21}, & \lambda_2 &= i(E_{21} - E_{12}), & \lambda_3 &= E_{11} - E_{22} \\ \lambda_4 &= E_{13} + E_{31}, & \lambda_5 &= i(E_{31} - E_{13}), & \lambda_6 &= E_{23} + E_{32} \\ \lambda_7 &= i(E_{32} - E_{23}), & \lambda_8 &= \frac{1}{\sqrt{3}}(E_{11} + E_{22} - 2E_{33}) \end{aligned}$$

With the identity

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{il}E_{kj} \quad (6)$$

$$\begin{aligned}
f_{123} &= \frac{1}{4i} \text{tr}([\lambda_1, \lambda_2] \lambda_3) = \frac{1}{4i} \text{tr}([E_{12} + E_{21}, i(E_{21} - E_{12})](E_{11} - E_{22})) \\
&= \frac{1}{4i} \text{tr}(i([E_{12}, E_{21}] - [E_{12}, E_{12}] + [E_{21}, E_{21}] - [E_{21}, E_{12}])(E_{11} - E_{22})) \\
&= \frac{1}{4i} \text{tr}(i(E_{11} - E_{22} - E_{22} + E_{11})(E_{11} - E_{22})) \\
&= \frac{1}{4i} \text{tr}(2i(E_{11} - E_{22})(E_{11} - E_{22})) \\
&= \frac{1}{4i} \text{tr}(2i(E_{11} + E_{22})) = \frac{1}{4i} 2i \cdot 2 = 1 \\
f_{123} &= 1
\end{aligned}$$

.

$$\begin{aligned}
f_{147} &= \frac{1}{4i} \text{tr}([\lambda_1, \lambda_4] \lambda_7) = \frac{1}{4i} \text{tr}([E_{12} + E_{21}, E_{13} + E_{31}] i(E_{32} - E_{23})) \\
&= \frac{1}{4i} \text{tr}(i([E_{12}, E_{13}] + [E_{12}, E_{31}] + [E_{21}, E_{13}] + [E_{21}, E_{31}])(E_{32} - E_{23})) \\
&= \frac{1}{4i} \text{tr}(i(-E_{32} + 0 + 0 + E_{23})(E_{32} - E_{23})) \\
&= \frac{1}{4i} \text{tr}(i(E_{33} + E_{22})) = \frac{1}{4i} i \cdot 2 = \frac{1}{2} \\
f_{147} &= \frac{1}{2}
\end{aligned}$$

.

$$\begin{aligned}
[\lambda_2, \lambda_4] &= [i(E_{21} - E_{12}), E_{13} + E_{31}] = i([E_{21}, E_{13}] + [E_{21}, E_{31}] - [E_{12}, E_{13}] - [E_{12}, E_{31}]) \\
&= i(0 + E_{23} + E_{13} + 0) = i(E_{23} + E_{32}) = i\lambda_6 \\
[\lambda_2, \lambda_4] &= 2if_{246}\lambda_6 \implies f_{246} = \frac{1}{2}
\end{aligned}$$

.

$$\begin{aligned}
[\lambda_2, \lambda_5] &= [i(E_{21} - E_{12}), i(E_{31} - E_{13})] = -([E_{21}, E_{31}] - [E_{21}, E_{13}] - [E_{12}, E_{31}] + [E_{12}, E_{13}]) \\
&= (0 + E_{23} - 0 - E_{32}) = (E_{23} - E_{32}) = i\lambda_7 \\
[\lambda_2, \lambda_5] &= 2if_{257}\lambda_7 \implies f_{257} = \frac{1}{2}
\end{aligned}$$

.

$$\begin{aligned}
[\lambda_3, \lambda_4] &= [E_{11} - E_{22}, E_{13} + E_{31}] = ([E_{11}, E_{13}] + [E_{11}, E_{31}] - [E_{22}, E_{13}] - [E_{22}, E_{31}]) \\
&= (E_{13} + 0 + 0 - E_{31}) = E_{13} - E_{31} = i\lambda_5 = 2if_{345}\lambda_5 \\
\implies f_{345} &= \frac{1}{2}.
\end{aligned}$$

$$\begin{aligned}
[\lambda_5, \lambda_1] &= [i(E_{31} - E_{13}), E_{12} + E_{21}] = i([E_{31}, E_{12}] + [E_{31}, E_{21}] - [E_{13}, E_{12}] - [E_{13}, E_{21}]) \\
&= i(E_{32} + 0 + 0 + E_{23}) = i(E_{32} + E_{23}) = i\lambda_6 \\
[\lambda_5, \lambda_1] &= 2if_{516}\lambda_6 \implies f_{516} = \frac{1}{2}.
\end{aligned}$$

$$\begin{aligned}
[\lambda_6, \lambda_3] &= [E_{23} + E_{32}, E_{11} - E_{22}] = ([E_{23}, E_{11}] - [E_{23}, E_{22}] + [E_{32}, E_{11}] - [E_{32}, E_{22}]) \\
&= (0 + E_{23} + 0 - E_{32}) = E_{23} - E_{32} = i\lambda_7 \\
[\lambda_6, \lambda_3] &= 2if_{637}\lambda_7 \implies f_{637} = \frac{1}{2}.
\end{aligned}$$

$$\begin{aligned}
[\lambda_4, \lambda_5] &= i [E_{13} + E_{31}, E_{31} - E_{13}] = i ([E_{13}, E_{31}] - [E_{13}, E_{13}] + [E_{31}, E_{31}] - [E_{31}, E_{13}]) \\
&= i (E_{11} - E_{33} - E_{33} + E_{11}) = 2i (E_{11} - E_{33}) \\
f_{JIL} &= \frac{1}{4i} \text{tr} ([\lambda_J, \lambda_I] \lambda_L) \rightarrow \frac{1}{4i} \text{tr} (2i (E_{11} - E_{33}) \lambda_8) \\
&= \frac{1}{4i} \text{tr} \left(2i (E_{11} - E_{33}) \frac{1}{\sqrt{3}} (E_{11} + E_{22} - 2E_{33}) \right) \\
f_{458} &= \frac{1}{4i} \text{tr} \left(\frac{1}{\sqrt{3}} (2iE_{11} + 4iE_{33}) \right) = \frac{1}{4i} \frac{6i}{\sqrt{3}} = 2\sqrt{3}
\end{aligned}$$

$$\begin{aligned}
[\lambda_6, \lambda_7] &= [E_{23} + E_{32}, i (E_{32} - E_{23})] = i ([E_{23}, E_{32}] - [E_{23}, E_{23}] + [E_{32}, E_{32}] - [E_{32}, E_{23}]) \\
&= i (E_{22} - E_{33} - E_{33} + E_{22}) = 2i (E_{22} - E_{33}) \\
f_{JIL} &= \frac{1}{4i} \text{tr} ([\lambda_J, \lambda_I] \lambda_L) \rightarrow \frac{1}{4i} \text{tr} (2i (E_{22} - E_{33}) \lambda_8) \\
&= \frac{1}{4i} \text{tr} \left(2i (E_{22} - E_{33}) \frac{1}{\sqrt{3}} (E_{11} + E_{22} - 2E_{33}) \right) \\
f_{678} &= \frac{1}{4i} \text{tr} \left(\frac{1}{\sqrt{3}} (2iE_{22} + 4iE_{33}) \right) = \frac{1}{4i} \frac{6i}{\sqrt{3}} = 2\sqrt{3}.
\end{aligned}$$

Combining the results we have the non-zero structure constants as

$$\begin{aligned}
f_{123} &= 1 \\
f_{147} = f_{246} = f_{257} = f_{345} = f_{516} = f_{637} &= \frac{1}{2} \\
f_{458} = f_{678} &= \frac{\sqrt{3}}{2}
\end{aligned}$$

Part (c)

$$\begin{aligned}
\{\lambda_I, \lambda_J\} &= \rho \delta_{IJ} \mathbf{1}_3 + 2d_{IJK} \lambda^K \\
\{\lambda_I, \lambda_J\} \lambda_L &= \rho \delta_{IJ} \mathbf{1}_3 + 2d_{IJK} \lambda^K \lambda_L \\
\text{tr} (\{\lambda_I, \lambda_J\} \lambda_L) &= \text{tr} (\rho \delta_{IJ} \mathbf{1}_3 \lambda_L + 2d_{IJK} \lambda^K \lambda_L) = \text{tr} (2d_{IJK} \lambda^K \lambda_L) \\
&= 2d_{IJK} 2\delta_{KL} = 4d_{IJL} \\
d_{IJL} &= \frac{1}{4} \text{tr} (\{\lambda_I, \lambda_J\} \lambda_L) = \frac{1}{4} \text{tr} (\lambda_I \lambda_J \lambda_L + \lambda_J \lambda_I \lambda_L)
\end{aligned}$$

Symmetry under $I \leftrightarrow J$:

$$d_{JIL} = \frac{1}{4} \text{tr} (\{\lambda_J, \lambda_I\} \lambda_L) = \frac{1}{4} \text{tr} (\lambda_J \lambda_I \lambda_L + \lambda_I \lambda_J \lambda_L) = d_{IJL}$$

To find the non-zero components we can use the same 2×2 subspaces $\mathcal{S}_{ij}$ as in part (b). Now we have to look at the anticommutator relations. We expect extra contributions as there are more degrees of freedom

in the symmetric case than the antisymmetric case.

We have the identity

$$\{E_{ij}, E_{kl}\} = \delta_{jk}E_{il} + \delta_{il}E_{kj} \quad (7)$$

$$\begin{aligned} \{\lambda_1, \lambda_1\} &= 2(E_{12} + E_{21})(E_{12} + E_{21}) = 2(E_{11} + E_{22}) \\ d_{11L} &= \frac{1}{4} \text{tr} (2(E_{11} + E_{22})\lambda_L) \rightarrow \frac{1}{4} \text{tr} \left(2(E_{11} + E_{22}) \frac{1}{\sqrt{3}} (E_{11} + E_{22} - 2E_{33}) \right) \\ &= \frac{1}{4} \text{tr} \left(\frac{2}{\sqrt{3}} (E_{11} + E_{22}) \right) = \frac{1}{4} \frac{4}{\sqrt{3}} = \frac{1}{\sqrt{3}} \\ d_{118} &= \frac{1}{\sqrt{3}} \end{aligned}$$

$$\begin{aligned} \{\lambda_1, \lambda_4\} &= \{E_{12} + E_{21}, E_{13} + E_{31}\} = \{E_{12}, E_{13}\} + \{E_{12}, E_{31}\} + \{E_{21}, E_{13}\} + \{E_{21}, E_{31}\} \\ &= E_{23} + E_{32} = \lambda_6 \\ d_{146} &= \frac{1}{4} \text{tr} (\lambda_6 \lambda_6) = \frac{1}{4} 2 = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \{\lambda_1, \lambda_5\} &= \{E_{12} + E_{21}, i(E_{31} - E_{13})\} = i(\{E_{12}, E_{31}\} - \{E_{12}, E_{13}\} + \{E_{21}, E_{31}\} - \{E_{21}, E_{13}\}) \\ &= i(E_{32} - E_{23}) = \lambda_7 \\ \{\lambda_1, \lambda_5\} &= 2d_{157}\lambda_7 \implies d_{157} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \{\lambda_2, \lambda_2\} &= \{i(E_{21} - E_{12}), i(E_{21} - E_{12})\} = -(\{E_{21}, E_{21}\} - \{E_{21}, E_{12}\} - \{E_{12}, E_{21}\} + \{E_{12}, E_{12}\}) \\ &= 2(E_{11} + E_{22}) = \frac{4}{3} \mathbf{1}_3 + 2d_{228}\lambda_8 = \frac{4}{3} [E_{11} + E_{22} + E_{33}] + 2d_{228} \left[\frac{1}{\sqrt{3}} (E_{11} + E_{22} - 2E_{33}) \right] \end{aligned}$$

Comparing independent coefficients

$$d_{228} = \frac{1}{\sqrt{3}}$$

The technique of comparing independent coefficients will be repeated and not typed up in latex.

$$\begin{aligned} \{\lambda_2, \lambda_4\} &= \{i(E_{21} - E_{12}), E_{13} + E_{31}\} = i(\{E_{21}, E_{13}\} + \{E_{21}, E_{31}\} - \{E_{12}, E_{13}\} - \{E_{12}, E_{31}\}) \\ &= i(E_{23} - E_{32}) = -\lambda_7 \\ \{\lambda_2, \lambda_4\} &= 2d_{247}\lambda_7 \implies d_{247} = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} \{\lambda_2, \lambda_5\} &= \{i(E_{21} - E_{12}), i(E_{31} - E_{13})\} = -(\{E_{21}, E_{31}\} - \{E_{21}, E_{13}\} - \{E_{12}, E_{31}\} + \{E_{12}, E_{13}\}) \\ &= E_{32} + E_{23} = \lambda_6 \\ \{\lambda_2, \lambda_5\} &= 2d_{256}\lambda_6 \implies d_{256} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned}\{\lambda_3, \lambda_3\} &= \{E_{11} - E_{22}, E_{11} - E_{22}\} = \{E_{11}, E_{11}\} - \{E_{11}, E_{22}\} - \{E_{22}, E_{11}\} + \{E_{22}, E_{22}\} \\ &= 2(E_{11} + E_{22})\end{aligned}$$

$$\{\lambda_3, \lambda_3\} = \frac{4}{3}\mathbf{1}_3 + 2d_{338}\lambda_8$$

$$d_{338} = \frac{1}{\sqrt{3}}$$

.

$$\begin{aligned}\{\lambda_3, \lambda_4\} &= \{E_{11} - E_{22}, E_{13} + E_{31}\} = \{E_{11}, E_{13}\} + \{E_{11}, E_{31}\} - \{E_{22}, E_{13}\} - \{E_{22}, E_{31}\} \\ &= E_{13} + E_{31} = \lambda_4\end{aligned}$$

$$\{\lambda_3, \lambda_4\} = 2d_{344}\lambda_4$$

$$d_{344} = \frac{1}{2}$$

.

$$\begin{aligned}\{\lambda_3, \lambda_5\} &= \{E_{11} - E_{22}, i(E_{31} - E_{13})\} = i(\{E_{11}, E_{31}\} - \{E_{11}, E_{13}\} - \{E_{22}, E_{31}\} + \{E_{22}, E_{13}\}) \\ &= i(E_{31} - E_{13}) = \lambda_5\end{aligned}$$

$$\{\lambda_3, \lambda_5\} = 2d_{355}\lambda_5$$

$$d_{355} = \frac{1}{2}$$

.

$$\begin{aligned}\{\lambda_3, \lambda_6\} &= \{E_{11} - E_{22}, E_{23} + E_{32}\} = \{E_{11}, E_{23}\} + \{E_{11}, E_{32}\} - \{E_{22}, E_{23}\} - \{E_{22}, E_{32}\} \\ &= -E_{23} - E_{32} = -\lambda_6\end{aligned}$$

$$\{\lambda_3, \lambda_6\} = 2d_{366}\lambda_6$$

$$d_{366} = -\frac{1}{2}$$

.

$$\begin{aligned}\{\lambda_3, \lambda_7\} &= \{E_{11} - E_{22}, i(E_{32} - E_{23})\} = i(\{E_{11}, E_{32}\} - \{E_{11}, E_{23}\} - \{E_{22}, E_{32}\} + \{E_{22}, E_{23}\}) \\ &= i(-E_{32} + E_{23}) = -\lambda_7\end{aligned}$$

$$\{\lambda_3, \lambda_7\} = 2d_{377}\lambda_7$$

$$d_{377} = -\frac{1}{2}$$

.

$$\begin{aligned}\{\lambda_4, \lambda_4\} &= \{E_{13} + E_{31}, E_{13} + E_{31}\} = \{E_{13}, E_{13}\} + \{E_{13}, E_{31}\} + \{E_{31}, E_{13}\} + \{E_{31}, E_{31}\} \\ &= 2(E_{11} + E_{33})\end{aligned}$$

$$\{\lambda_4, \lambda_4\} = \frac{4}{3}\mathbf{1}_3 + 2d_{448}\lambda_8$$

$$d_{448} = -\frac{1}{2\sqrt{3}}$$

.

$$\begin{aligned}\{\lambda_5, \lambda_5\} &= \{i(E_{31} - E_{13}), i(E_{31} - E_{13})\} = -(\{E_{31}, E_{31}\} - \{E_{31}, E_{13}\} - \{E_{13}, E_{31}\} + \{E_{13}, E_{13}\}) \\ &= 2(E_{11} + E_{33})\end{aligned}$$

$$\{\lambda_5, \lambda_5\} = \frac{4}{3}\mathbf{1}_3 + 2d_{558}\lambda_8$$

$$d_{558} = -\frac{1}{2\sqrt{3}}$$

.

$$\begin{aligned}\{\lambda_6, \lambda_6\} &= \{E_{23} + E_{32}, E_{23} + E_{32}\} = \{E_{23}, E_{23}\} + \{E_{23}, E_{32}\} + \{E_{32}, E_{23}\} + \{E_{32}, E_{32}\} \\ &= 2(E_{22} + E_{33})\end{aligned}$$

$$\{\lambda_6, \lambda_6\} = \frac{4}{3}\mathbf{1}_3 + 2d_{668}\lambda_8$$

$$d_{668} = -\frac{1}{2\sqrt{3}}$$

.

$$\begin{aligned}\{\lambda_7, \lambda_7\} &= \{i(E_{32} - E_{23}), i(E_{32} - E_{23})\} = -(\{E_{32}, E_{32}\} - \{E_{32}, E_{23}\} - \{E_{23}, E_{32}\} + \{E_{23}, E_{23}\}) \\ &= 2(E_{22} + E_{33})\end{aligned}$$

$$\{\lambda_7, \lambda_7\} = \frac{4}{3}\mathbf{1}_3 + 2d_{778}\lambda_8$$

$$d_{778} = -\frac{1}{2\sqrt{3}}$$

.

$$\begin{aligned}\{\lambda_8, \lambda_8\} &= \left\{ \frac{1}{\sqrt{3}}(E_{11} + E_{22} - 2E_{33}), \frac{1}{\sqrt{3}}(E_{11} + E_{22} - 2E_{33}) \right\} \\ &= \frac{1}{3}(\{E_{11}, E_{11}\} + \{E_{11}, E_{22}\} - 2\{E_{11}, E_{33}\} + \{E_{22}, E_{11}\}) \\ &\quad + \frac{1}{3}(\{E_{22}, E_{22}\} - 2\{E_{22}, E_{33}\} - 2\{E_{33}, E_{11}\} - 2\{E_{33}, E_{22}\} + 4\{E_{33}, E_{33}\}) \\ &= \frac{1}{3}(2E_{11} + 2E_{22} + 4E_{33}) = \frac{2}{3}\mathbf{1}_3\end{aligned}$$

$$\{\lambda_8, \lambda_8\} = \frac{4}{3}\mathbf{1}_3 + 2d_{888}\lambda_8$$

$$d_{888} = -\frac{1}{\sqrt{3}}$$

.

Combining the results we have the non-zero symmetric structure constants as

$$\begin{aligned}
d_{118} &= d_{228} = d_{338} = \frac{1}{\sqrt{3}} \\
d_{448} &= d_{558} = d_{668} = d_{778} = -\frac{1}{2\sqrt{3}} \\
d_{888} &= -\frac{1}{\sqrt{3}} \\
d_{157} &= d_{146} = d_{256} = d_{344} = d_{355} = \frac{1}{2} \\
d_{366} &= d_{377} = d_{247} = -\frac{1}{2}
\end{aligned}$$

Part (d)

The quadratic Casimir operator is defined as

$$C_2 = \sum_{I=1}^8 \lambda_I \lambda_I \quad (8)$$

We can use the 2×2 subspaces $\mathcal{S}_{ij}$ as in part (b) to evaluate the sum. Each subspace will contribute a term proportional to the identity matrix in that subspace. Summing over all subspaces will give a term proportional to the identity matrix in the full 3×3 space.

In the subspace $\mathcal{S}_{12}$ we have the contributions from $\lambda_1, \lambda_2, \lambda_3$:

$$\begin{aligned}
\lambda_1^2 &= (E_{12} + E_{21})(E_{12} + E_{21}) = E_{11} + E_{22} \\
\lambda_2^2 &= (i(E_{21} - E_{12}))^2 = -(E_{21} - E_{12})(E_{21} - E_{12}) = E_{11} + E_{22} \\
\lambda_3^2 &= (E_{11} - E_{22})^2 = E_{11} + E_{22}
\end{aligned}$$

Summing these we get

$$\lambda_1^2 + \lambda_2^2 + \lambda_3^2 = 3(E_{11} + E_{22}) \quad (9)$$

In the subspace $\mathcal{S}_{13}$ we have the contributions from $\lambda_4, \lambda_5, \lambda_8$:

$$\begin{aligned}
\lambda_4^2 &= (E_{13} + E_{31})(E_{13} + E_{31}) = E_{11} + E_{33} \\
\lambda_5^2 &= (i(E_{31} - E_{13}))^2 = -(E_{31} - E_{13})(E_{31} - E_{13}) = E_{11} + E_{33} \\
\lambda_8^2 &= \left(\frac{1}{\sqrt{3}} (E_{11} + E_{22} - 2E_{33}) \right)^2 = \frac{1}{3} (E_{11} + E_{22} + 4E_{33})
\end{aligned}$$

Summing these we get

$$\lambda_4^2 + \lambda_5^2 + \lambda_8^2 = 2(E_{11} + E_{33}) + \frac{1}{3}(E_{11} + E_{22} + 4E_{33}) = \frac{7}{3}E_{11} + \frac{1}{3}E_{22} + \frac{10}{3}E_{33} \quad (10)$$

In the subspace $\mathcal{S}_{23}$ we have the contributions from $\lambda_6, \lambda_7, \lambda_8$:

$$\begin{aligned}
\lambda_6^2 &= (E_{23} + E_{32})(E_{23} + E_{32}) = E_{22} + E_{33} \\
\lambda_7^2 &= (i(E_{32} - E_{23}))^2 = -(E_{32} - E_{23})(E_{32} - E_{23}) = E_{22} + E_{33} \\
\lambda_8^2 &= \left(\frac{1}{\sqrt{3}} (E_{11} + E_{22} - 2E_{33}) \right)^2 = \frac{1}{3} (E_{11} + E_{22} + 4E_{33})
\end{aligned}$$

Summing these we get

$$\lambda_6^2 + \lambda_7^2 + \lambda_8^2 = 2(E_{22} + E_{33}) + \frac{1}{3}(E_{11} + E_{22} + 4E_{33}) = \frac{1}{3}E_{11} + \frac{7}{3}E_{22} + \frac{10}{3}E_{33} \quad (11)$$

Adding all contributions from the three subspaces we have

$$\begin{aligned} C_2 &= \sum_{I=1}^8 \lambda_I \lambda_I = (\lambda_1^2 + \lambda_2^2 + \lambda_3^2) + (\lambda_4^2 + \lambda_5^2) + (\lambda_6^2 + \lambda_7^2 + \lambda_8^2) \\ &= 3(E_{11} + E_{22}) + 2(E_{11} + E_{33}) + \frac{1}{3}E_{11} + \frac{7}{3}E_{22} + \frac{10}{3}E_{33} \\ &= \left(3 + 2 + \frac{1}{3}\right)E_{11} + \left(3 + \frac{7}{3}\right)E_{22} + \left(2 + \frac{10}{3}\right)E_{33} = \frac{11}{3}E_{11} + \frac{16}{3}E_{22} + \frac{16}{3}E_{33} \\ &= \frac{16}{3}(E_{11} + E_{22} + E_{33}) = \frac{16}{3}\mathbf{1}_3 \end{aligned}$$

Thus the quadratic Casimir operator in the fundamental representation is

$$C_2 = \frac{16}{3}\mathbf{1}_3 = \frac{16}{3}[E_{11} + E_{22} + E_{33}] \quad (12)$$

For the cubic Casimir operator we have

$$\tilde{C}_3 = d_{IJK}\lambda_I\lambda_J\lambda_K \quad (13)$$

Where we can identify $E_I = i\lambda_I$.

By definition the cubic Casimir operator must commute with all generators of the algebra. Thus it must be proportional to the identity matrix in the fundamental representation. Therefore we can write

$$C_3 = c\mathbf{1}_3 \quad (14)$$

for some constant c .

$$\begin{aligned} \lambda_I\lambda_J &= \frac{1}{2}(\{\lambda_I, \lambda_J\} + [\lambda_I, \lambda_J]) = \frac{1}{2}\left(\frac{4}{3}\delta_{IJ}\mathbf{1}_3 + 2d_{IJK}\lambda_K + 2if_{IJK}\lambda_K\right) \\ tr(\lambda_I\lambda_J\lambda_K) &= \frac{1}{2}tr\left(\left(\frac{4}{3}\delta_{IJ}\mathbf{1}_3 + 2d_{IJK}\lambda_K + 2if_{IJK}\lambda_K\right)\lambda_K\right) \\ &= \frac{1}{2}tr\left(\frac{4}{3}\delta_{IJ}\lambda_K + 2d_{IJK}\lambda_K\lambda_K + 2if_{IJK}\lambda_K\lambda_K\right) \\ &= \frac{1}{2}\left(\frac{4}{3}\delta_{IJ}tr(\lambda_K) + 2d_{IJK}tr(\lambda_K\lambda_K) + 2if_{IJK}tr(\lambda_K\lambda_K)\right) \\ tr(\lambda_I\lambda_J\lambda_K) &= \frac{1}{2}(0 + 2d_{IJK}2\delta_{LK} + 2if_{IJK}2\delta_{LK}) = 2d_{IJK} + 2if_{IJK} \end{aligned}$$

$$\begin{aligned} tr\tilde{C}_3 &= tr(d_{IJK}\lambda_I\lambda_J\lambda_K) = d_{IJK}tr(\lambda_I\lambda_J\lambda_K) = d_{IJK}(2d_{IJK} + 2if_{IJK}) \\ &= 2d_{IJK}d_{IJK} \\ tr\tilde{C}_3 &= tr(c\mathbf{1}_3) = 3c \\ \implies c &= \frac{2}{3}d_{IJK}d_{IJK}. \end{aligned}$$

Thus we need to sum over the symmetric structure constants (not just the independent ones) to find the value of c .

First lets sum the independent non-zero components squared:

$$\begin{aligned}
S &= d_{118}^2 + d_{228}^2 + d_{338}^2 + d_{448}^2 + d_{558}^2 + d_{668}^2 + d_{778}^2 + d_{888}^2 + d_{146}^2 + d_{157}^2 + d_{247}^2 + d_{256}^2 + d_{344}^2 + d_{355}^2 + d_{366}^2 + d_{377}^2 \\
&= 3 \left(\frac{1}{\sqrt{3}} \right)^2 + 4 \left(-\frac{1}{2\sqrt{3}} \right)^2 + \left(-\frac{1}{\sqrt{3}} \right)^2 + 4 \left(\frac{1}{2} \right)^2 + 2 \left(-\frac{1}{2} \right)^2 \\
&= \frac{3}{3} + 4 \frac{1}{12} + \frac{1}{3} + 4 \frac{1}{4} + 2 \frac{1}{4} = 1 + \frac{1}{3} + \frac{1}{3} + 1 + \frac{1}{2} = \frac{11}{3}
\end{aligned}$$

Now we need to account for the multiplicities due to the symmetry of the indices.

For terms with distinct indices we have $3! = 6$ permutations. For terms with two indices equal and one different we have 3 permutations. For terms with all indices equal there is only 1 permutation.

Thus the total sum is

$$\begin{aligned}
\sum_{I,J,K=1}^8 d_{IJK} d_{IJK} &= 6 (d_{146}^2 + d_{157}^2 + d_{247}^2 + d_{256}^2) + 3 (d_{118}^2 + d_{228}^2 + d_{338}^2 + d_{448}^2 + d_{558}^2 + d_{668}^2 + d_{778}^2) + d_{888}^2 \\
&\quad + 3 (d_{344}^2 + d_{355}^2 + d_{366}^2 + d_{377}^2) \\
&= 6 \left(\frac{1}{2}^2 + \frac{1}{2}^2 + \left(-\frac{1}{2} \right)^2 + \frac{1}{2}^2 \right) + 3 \left(\left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 + 4 \left(-\frac{1}{2\sqrt{3}} \right)^2 \right) + \left(-\frac{1}{\sqrt{3}} \right)^2 \\
&\quad + 3 \left(\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 + \left(-\frac{1}{2} \right)^2 + \left(-\frac{1}{2} \right)^2 \right) \\
&= 6 \left(4 \frac{1}{4} \right) + 3 \left(3 \frac{1}{3} + 4 \frac{1}{12} \right) + \frac{1}{3} + 3 = 6 + 3 \left(1 + \frac{1}{3} \right) + \frac{1}{3} + 3 = 6 + 4 + \frac{1}{3} + 3 = \frac{40}{3}
\end{aligned}$$

$$\begin{aligned}
tr \tilde{C}_3 &= tr 3c = tr 3 \left(\frac{2}{3} d_{IJK} d_{IJK} \right) = 3 \left[\frac{2}{3} \left(\frac{40}{3} \right) \right] = \frac{80}{3} \\
3c &= \frac{80}{3} \implies c = \frac{80}{9} \\
d_{IJK} \lambda_I \lambda_J \lambda_K &= \frac{80}{9} \mathbf{1}_3
\end{aligned}$$

Using $E_I = i\lambda_I$

$$C_3 = i^3 d_{IJK} \lambda_I \lambda_J \lambda_K = -i \frac{80}{9} \mathbf{1}_3 \quad (15)$$

In terms of E_{ij}

$$C_3 = -i \frac{80}{9} (E_{11} + E_{22} + E_{33}) \quad (16)$$

Part e

$$\begin{aligned}
[E_I, E_J] &= [i\lambda_I, i\lambda_J] = -2f_{IJK} \\
ad_Z(E_J) &= [Z, E_J] = -2z_I f_{IJK} E_K \\
(ad_Z)_{KJ} &= -2 \sum_{I=1}^8 z_I f_{IJK}.
\end{aligned}$$

$$f_{123} = 1, \quad f_{147} = f_{246} = f_{257} = f_{345} = f_{516} = f_{637} = \frac{1}{2} \quad f_{458} = f_{678} = \frac{\sqrt{3}}{2}.$$

$$\begin{aligned} [Z, E_1] &= 2z_2E_3 - 2z_3E_2 + z_4E_7 - z_5E_6 + z_6E_5 - z_7E_4, \\ [Z, E_2] &= 2z_3E_1 - 2z_1E_3 + z_4E_6 - z_6E_4 + z_5E_7 - z_7E_5, \\ [Z, E_3] &= 2z_1E_2 - 2z_2E_1 + z_4E_5 - z_5E_4 + z_6E_7 - z_7E_6, \\ [Z, E_4] &= z_7E_1 + z_6E_2 + z_5E_3 - z_2E_6 - z_1E_7 - (z_3 + \sqrt{3}z_8)E_5 + \sqrt{3}z_5E_8, \\ [Z, E_5] &= z_6E_1 + z_7E_2 - z_4E_3 + (z_3 + \sqrt{3}z_8)E_4 - z_1E_6 - z_2E_7 - \sqrt{3}z_4E_8, \\ [Z, E_6] &= -z_5E_1 - z_4E_2 + z_7E_3 + z_2E_4 + z_1E_5 - (z_3 + \sqrt{3}z_8)E_7 + \sqrt{3}z_7E_8, \\ [Z, E_7] &= -z_4E_1 - z_5E_2 - z_6E_3 + z_1E_4 + z_2E_5 + (z_3 + \sqrt{3}z_8)E_6 - \sqrt{3}z_6E_8, \\ [Z, E_8] &= -\sqrt{3}z_5E_4 + \sqrt{3}z_4E_5 - \sqrt{3}z_7E_6 + \sqrt{3}z_6E_7. \end{aligned}$$

$$A_{KJ} = \begin{bmatrix} 0 & 2z_3 & -2z_2 & 0 & 0 & 0 & 0 & 0 \\ -2z_3 & 0 & 2z_1 & 0 & 0 & 0 & 0 & 0 \\ 2z_2 & -2z_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -z_7 & 0 & 0 & 0 & z_5 & 0 & -z_1 & 0 \\ z_6 & 0 & 0 & -z_5 & 0 & -z_1 & 0 & -\sqrt{3}z_4 \\ -z_5 & z_4 & 0 & -z_2 & z_1 & 0 & -z_3 - \sqrt{3}z_8 & 0 \\ z_4 & -z_5 & -z_3 & z_1 & z_2 & z_3 + \sqrt{3}z_8 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{3}z_5 & -\sqrt{3}z_4 & \sqrt{3}z_7 & -\sqrt{3}z_6 & 0 \end{bmatrix} \quad (17)$$

Part f

$$\begin{aligned} (ad_Z)_{JM}^2 &= 4z_I z_L f_{IJK} f_{LKM} \\ K(Z, Z) &= tr(ad_Z^2) = 4z_I z_L f_{IJK} f_{LKM} \\ f_{IJK} f_{LJK} &= C_A \delta_{IL} \\ C_A &= N = 3 \\ K(Z, Z) &= -4 \cdot 3z_I z_I = -12\vec{z} \cdot \vec{z} \\ K(Z, Z) &= tr Z^2 = z_I z_J tr(E_I E_J) = -2z_I z_I = -2\vec{z} \cdot \vec{z}. \\ K(Z, Z) &= -12\vec{z}^2 \quad \tilde{K}(Z, Z) = -2\vec{z}^2 \end{aligned} \quad (18)$$

part e

$$\begin{aligned} \psi_3 &= d_{IJK} z_I z_J z_K \\ &= 3(z_1 z_4 z_6 + z_2 z_5 z_6 + z_1 z_5 z_7 - z_2 z_5 z_7 - z_2 z_4 z_7) + \frac{3}{2} z_3 (z_4^2 + z_5^2 - z_6^2 - z_7^2) \\ &\quad + \frac{\sqrt{3}}{2} z_8 (2z_1^2 + 2z_2^2 - z_4^2 - z_5^2 - z_6^2 - z_7^2) + \sqrt{3} z_3^2 z_8 - \frac{1}{\sqrt{3}} z_8^2 \\ &= \dots \end{aligned}$$

$$\begin{aligned}
p_X(\lambda) &= -\lambda^3 + \frac{1}{2} \text{tr} X^2 \lambda + \det X \\
\text{traceless } 3 \times 3 &\implies \det X = \frac{1}{3} \text{tr}(X^3) \\
\text{tr} Z^2 &= z_I z_J \text{tr}(E_I E_J) \\
&= -\text{tr}(\lambda_I \lambda_J) z_I z_J \\
&= -2\psi_2(\vec{z}) \\
\text{tr} Z^3 &= i^3 z_I z_J z_K \text{tr}(E_I E_J E_K) \\
&= -2i\psi_3(\vec{z}) \\
p_Z(\lambda) &= -\lambda^3 - \psi_2(\vec{z}) \lambda - \frac{2i}{3} \psi_3(\vec{z}).
\end{aligned}$$

part g

$$ad_Z(E_J) = -2(f_{3JK} z_3 + f_{8JK} z_8) E_K \quad (19)$$

The adjoint matrix now looks like

$$ad_Z = \begin{bmatrix} 0 & 2z_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2z_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & z_3 + \sqrt{3}z_8 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -z_3 - \sqrt{3}z_8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -z_3 + \sqrt{3}z_8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & z_3 - \sqrt{3}z_8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (20)$$

We have eigenvalues given by

$$\begin{aligned}
\lambda_1 &= \lambda_2 = 0 \\
\lambda_3 &= -\lambda_4 = 2iz_3 \\
\lambda_5 &= -\lambda_6 = -i(z_3 - \sqrt{3}z_8) \\
\lambda_7 &= -\lambda_8 = i(z_3 + \sqrt{3}z_8).
\end{aligned}$$

$$\begin{aligned}
X_{12}^+ &= E_1 + iE_2 = i(\lambda_1 + i\lambda_2) = 2i e_{12}, \\
X_{12}^- &= E_1 - iE_2 = i(\lambda_1 - i\lambda_2) = 2i e_{21}, \\
X_{13}^+ &= E_4 + iE_5 = i(\lambda_4 + i\lambda_5) = 2i e_{13}, \\
X_{13}^- &= E_4 - iE_5 = i(\lambda_4 - i\lambda_5) = 2i e_{31}, \\
X_{23}^+ &= E_6 + iE_7 = i(\lambda_6 + i\lambda_7) = 2i e_{23}, \\
X_{23}^- &= E_6 - iE_7 = i(\lambda_6 - i\lambda_7) = 2i e_{32}, \\
H_3 &= E_3 = i\lambda_3 = i \text{diag}(1, -1, 0), \\
H_8 &= E_8 = i\lambda_8 = \frac{i}{\sqrt{3}} \text{diag}(1, 1, -2)..
\end{aligned}$$

Eigenvalue $+2iz_3$:	$X_{12}^+ = 2i e_{12},$
Eigenvalue $-2iz_3$:	$X_{12}^- = 2i e_{21},$
Eigenvalue $+i(z_3 + \sqrt{3}z_8)$:	$X_{13}^+ = 2i e_{13},$
Eigenvalue $-i(z_3 + \sqrt{3}z_8)$:	$X_{13}^- = 2i e_{31},$
Eigenvalue $+i(z_3 - \sqrt{3}z_8)$:	$X_{23}^+ = 2i e_{23},$
Eigenvalue $-i(z_3 - \sqrt{3}z_8)$:	$X_{23}^- = 2i e_{32},$
Eigenvalue 0 :	$H_3 = i \text{diag}(1, -1, 0), \quad H_8 = \frac{i}{\sqrt{3}} \text{diag}(1, 1, -2)..$

Explicitly in terms of matrices we have

$$\begin{aligned}
X_{12}^+ &= c_{12}^+ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} & X_{12}^- &= c_{12}^- \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
X_{13}^+ &= c_{13}^+ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} & X_{13}^- &= c_{13}^- \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\
X_{23}^+ &= c_{23}^+ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} & X_{23}^- &= c_{23}^- \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\
H_3 &= \tilde{c}_3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} & H_8 &= \tilde{c}_8 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.
\end{aligned}$$